

ALARA

Proof of concept

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Executive Summary

Ground level air quality monitoring across the Gulf Cooperation Council (GCC) states faces a structural challenge: surface stations are geographically sparse relative to the region's dense industrial footprint and heavy oil and gas operations. Project ELARA bridges this gap by leveraging high-resolution satellite remote sensing to provide continuous, facility level emissions records without deploying physical hardware.

Using Sentinel-5P (TROPOMI) tropospheric column density measurements, ELARA transforms open-access Earth Observation data into actionable environmental intelligence. This document presents our atmospheric analysis, localized multi-decade pollution trends, automated technical methodology, and commercial viability strategy for hackathon judging.

1. Context & Atmospheric Science

The Ground Monitoring Gap

Traditional air monitoring relies on fixed ground stations. In vast industrial corridors or offshore platforms, this creates critical blind spots:

- Coverage Gaps: Significant unmonitored zones exist between static stations and over offshore extraction fields.
- Detection Lag: Ground sensors capture localized ambient conditions rather than identifying the precise source or direction of an advancing plume.
- High Infrastructure Costs: Physical stations cost tens of thousands of dollars per unit, limiting rapid regional scaling.

Primary Target Pollutants & Photochemistry:

Satellites measure the tropospheric column density—a vertical integration of molecules over a target area (molecules/cm² or $\mu\text{mol}/\text{m}^2$)—rather than single-point surface concentrations.

Compound	Formula	Primary Regional Sources	Environmental & Health Impact
Nitrogen Dioxide	NO ₂	Vehicle exhaust, power plants, industrial combustion	Respiratory irritant; primary precursor to ground-level ozone (O ₃) and smog.
Sulfur Dioxide	SO ₂	Heavy fuel oil combustion, refineries, marine shipping	Acid rain precursor; secondary driver of fine particulate matter (PM _{2.5})
Formaldehyde	HCHO	Petrochemical refining, incomplete combustion	Key marker of Volatile Organic Compound (VOC) activity and ozone formation.
Glyoxal	CHOCHO	Secondary photochemical reactions	Direct indicator of atmospheric oxidation capacity.
Particulate Matter	PM ₁₀ /PM _{2.5}	Industrial dust, combustion by products, construction	Deep respiratory hazard; modeled via surface reflectance where stations are lacking.

2. Quantified Regional Trends (2005–2023)

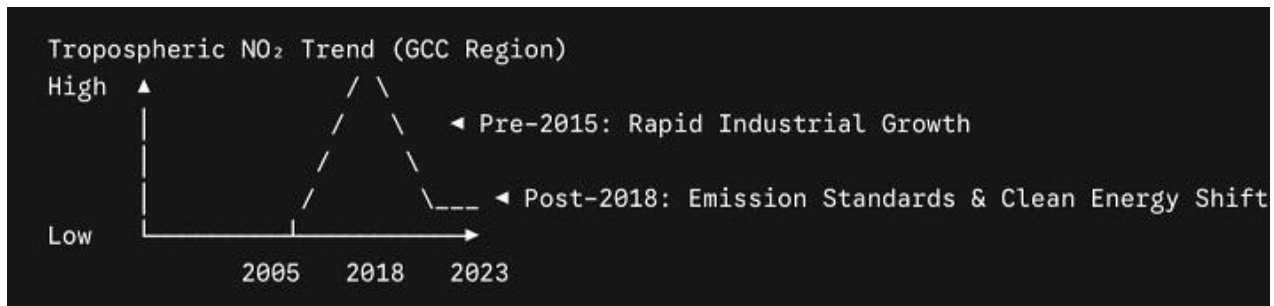
Consolidated findings from peer-reviewed literature and long-term OMI/TROPOMI observational data highlight two distinct atmospheric regimes over the Arabian Peninsula:

Middle East Pollution Trends

Region / Location Type	Pollutant	Quantified Trend	Timeframe	Primary Driver
Urban Areas (Middle East-wide)	NO ₂	Up to +12% / year	2005–2014	Rapid urbanization & rising vehicle ownership.
Refineries, Oil Ports, Power Plants	NO ₂	+2% to +9% / year	2005–2014	Expansion of energy infrastructure.
Industrial Hubs (Kuwait City, Dubai)	NO ₂	+	2005–2023	Sustained industrial throughput.
Qatar, Bahrain, Dammam Corridor	NO ₂	–1.7% / year	2018–2023	Tightened fuel standards & cleaner power generation.
Offshore Platforms (Gulf Comparison)	NO ₂	Up to +25% over background	2004–2022	Uncontrolled flaring & stationary power generation.
Industrial Hotspots (General)	SO ₂	+0.01 DU / year	2005–2023	High-sulfur fuel oil consumption.

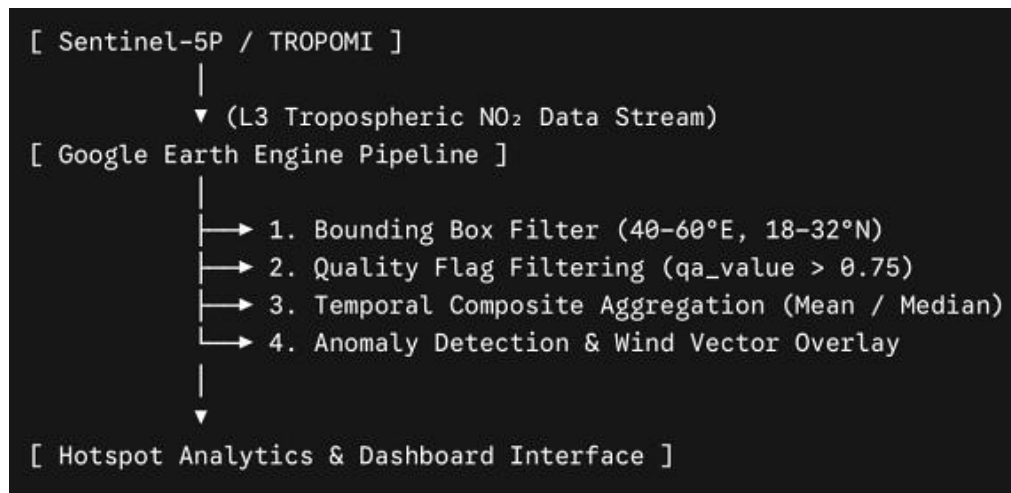
Statistical Turning Point

Emissions surged through the mid-2010s due to economic growth. However, starting around 2015–2018, policy interventions such as stricter vehicle emission rules, transition to natural gas, and renewable deployment—began reversing trends in major sub-regions.



3. Technical Methodology & Implementation

ELARA transitions from legacy OMI satellite data (resolution) to TROPOMI aboard Sentinel-5P (resolution). This fine spatial resolution enables pinpointing individual facility plumes rather than broad municipal averages.



Reproducible Google Earth Engine Script (Java script)

```
// ELARA: TROPOMI NO2 Hotspot Detection & Trend Analysis
var aoi = ee.Geometry.Rectangle([40.0, 18.0, 60.0, 32.0]);

// Load Sentinel-5P Tropospheric NO2 collection
var no2Collection = ee.ImageCollection('COPERNICUS/S5P/OFFL/L3_NO2')
  .select('tropospheric_NO2_column_number_density')
  .filterBounds(aoi);

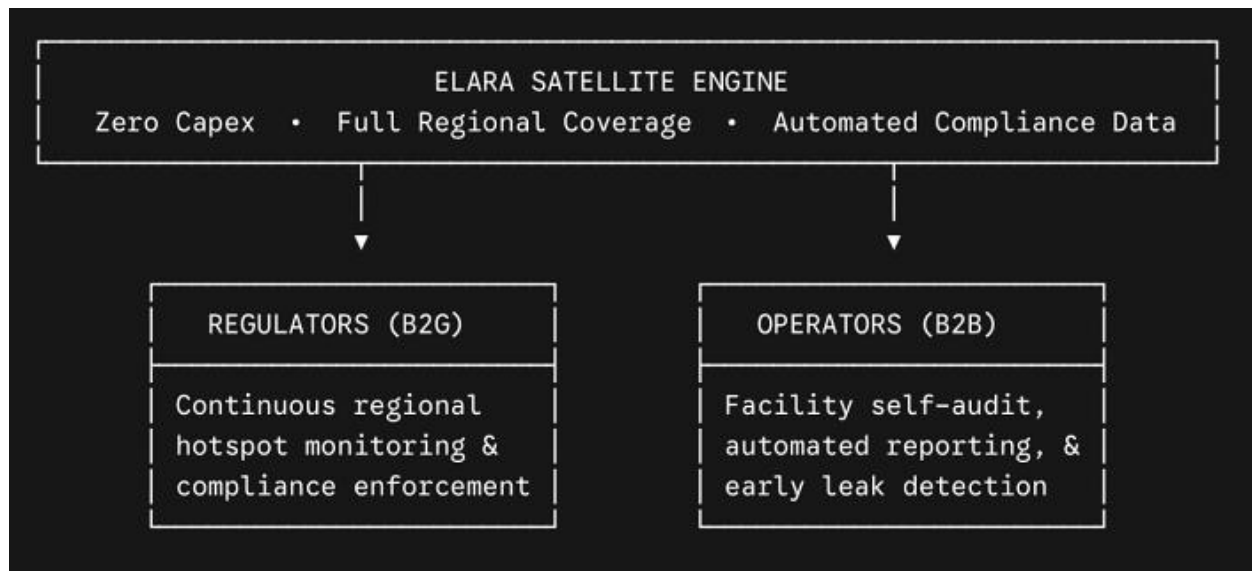
// Define baseline (Pre-intervention) and current period
var baseline = no2Collection.filterDate('2019-01-01', '2019-12-31').mean().clip(aoi);
var current = no2Collection.filterDate('2023-01-01', '2023-12-31').mean().clip(aoi);

// Calculate absolute percentage change across the AOI
var percentageChange = current.subtract(baseline).divide(baseline).multiply(100);

// Visualization parameters
var visParams = {
  min: 0.00002,
  max: 0.00015,
  palette: ['black', 'blue', 'purple', 'cyan', 'green', 'yellow', 'red']
};

Map.centerObject(aoi, 5);
Map.addLayer(baseline, visParams, 'NO2 Baseline (2019)');
Map.addLayer(current, visParams, 'NO2 Current (2023)');
Map.addLayer(percentageChange, {min: -20, max: 20, palette: ['blue', 'white', 'red']}, 'Percentage Change');
```

"Turning free satellite data into continuous, facility level emissions records without installing a single physical sensor."



Comparative Advantage

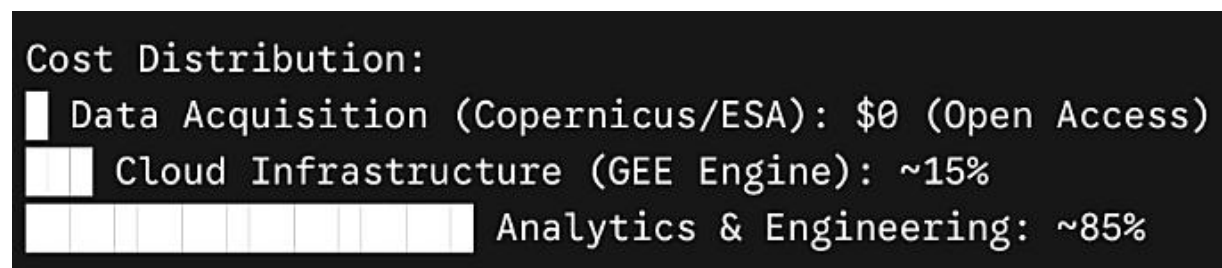
Approach	Coverage	Deployment Cost	Resolution / Specificity
Ground Stations	Highly localized ($\approx 1\text{--}2$ km radius)	High ($\approx \$30\text{k--}\100k per unit)	Point-source specific, but sparse.
Manual Audits	Periodic / Infrequent	High recurring labor costs	Self-reported snapshot; prone to gaps.
Generic AQ Apps	Regional / Municipal	Low	Broad municipal averages; no point attribution.
ELARA Platform	Full GCC regional coverage	Near-zero infrastructure Capex	Facility-level attribution (3.5×5.5 km)

Revenue Architecture & Business Model

1. Freemium Public Tier: Free regional air quality index maps to engage stakeholders and fulfill public transparency goals. B2G Government Subscriptions: Annual contracts for national environment agencies providing regional dashboards, anomaly alerts, and automated policy reports.

2. B2B Industrial Tier: Monthly/annual SaaS subscriptions for oil, gas, and industrial operators offering private facility monitoring, audit prep, and early plume detection.
3. On-Demand Industrial Audits: Pay-per-report historical emissions audits for ESG auditors, financial institutions, and insurers evaluating risk.

Lean Cost Structure



- **Data Costs: \$0.** Built entirely on open-access Copernicus Sentinel-5P data.
- **Compute Infrastructure:** Scalable cloud compute via Google Earth Engine, removing upfront hardware investment.
- **Primary Resource Allocation:** Focused on analytics, model validation, and user dashboard development.

6. Conclusion & Future Roadmap

Project ELARA demonstrates that satellite remote sensing offers an immediate, low-cost solution to the region's air quality monitoring gaps. By translating free, high-resolution Earth Observation data into continuous, facility-level emissions records, ELARA empowers both regulatory bodies and industrial operators to enforce standards, identify leaks early, and accelerate the Gulf's transition toward sustainable growth.

